

Task 3 :

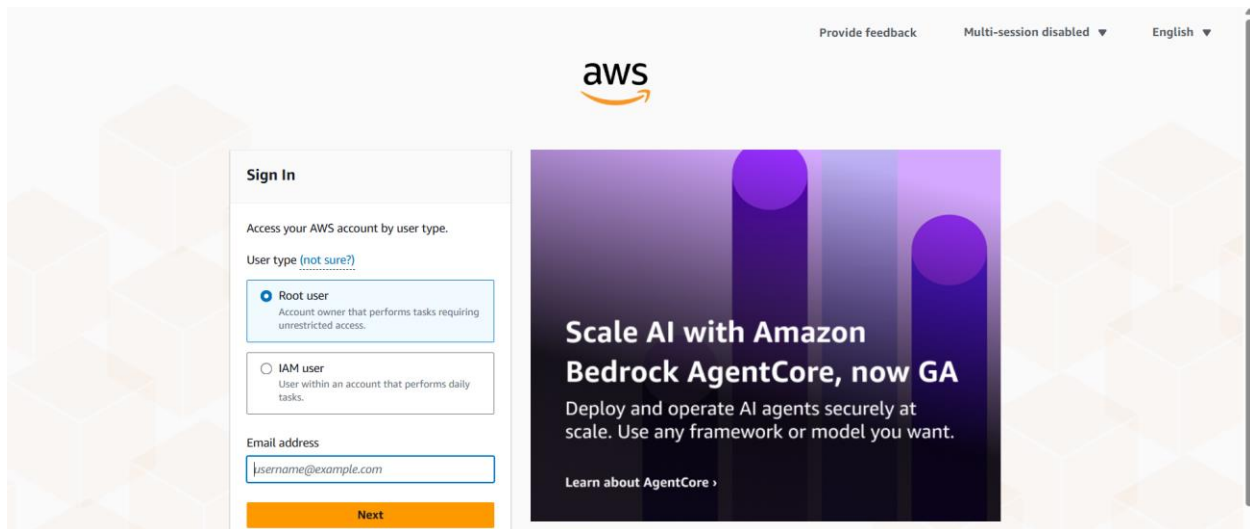
1. Create a **Private EC2 instance**.
2. Create a **Bastion Server (Public EC2)** and establish connectivity to the private EC2 through it.
3. Host a **live website on the private EC2**.
4. Configure a **Load Balancer** in such a way that:
 - a. The website is deployed on the **private EC2 (private subnet)**.
 - b. The website is accessible in a browser using the **Load Balancer's IP/DNS**.

Solution:

1. Login to your AWS account with credentials and required verification.

(<https://signin.aws.amazon.com/>)

My account – AWS Management Console- Sign in



2. Create a custom VPC:
 - Name – *abc*
 - Assign IPV4 CIDR Block – 10.0.0.0/16
 - (*CIDR block must be in between /16 to /24*)
 - Tendency – Default

VPC > Your VPCs > Create VPC

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

my-vpc

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
 10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)
 Default

VPC encryption control (\$) [Info](#)
 Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#)

☒ None
☐ Monitor mode
See which resources in your VPC are unencrypted but allow the creation of unencrypted resources.
☐ Enforce mode
Requires all resources, except exclusions, in your VPC to be encryption-capable and blocks creation of unencrypted resources.

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3. Create Public and Private Subnet

- Go to subnet – create subnet -
- Select existing VPC- Give name- Assign IPv4 subnet CIDR block to it (10.0.1.0/24 CIDR Block must be in between 16 to 24)
- Create Public Subnet

VPC > Subnets > Create subnet

VPC ID
Create subnets in this VPC.
 vpc-0ecb2db9c6922b0c (my-vpc)

Associated VPC CIDRs
 IPv4 CIDRs
 10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
 pb-subnet
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
 Europe (Stockholm) / eu-north-1a

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
 10.0.0.0/16

IPv4 subnet CIDR block

4. Create Public and Private Subnet

- Go to subnet – create subnet -
- Select existing VPC- Give name- Assign IPv4 subnet CIDR block to it. (10.0.2.0/24 CIDR Block must be in between 16 to 24)
- Create Private Subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

pv-subnet

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / us-east-1a

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.2.0/24 256 IPs

Tags - optional

Key	Value - optional	
Q Name	pv-subnet	(Remove)

Internet Gateway to and connect it to VPC.

1. Goto Internet Gateway
2. Create Gateway
3. Give name (*my-igw*)
4. Select launched IGW and attach to VPC

Attach to VPC (igw-0488a0d97ecbef5)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

Q vpc-05d8c76cbaeb9e259

► AWS Command Line Interface command

Cancel Attach internet gateway

6. Create Route table for public subnet

- go to route table
- Create route table
- Use existing VPC
- Go to edit routes add destination(**0.0.0.0/0**) and target(**Select Created internet gateway**) and save route.
- Associate subnet – edit subnet association – select public subnet and save.
- Then Enable Auto assign public Ip- go to subnet – select public subnet – edit subnet setting- then enable option of Auto assign public IP V4 address and save changes.

VPC > Route tables > rtb-065d968ea395aac31

Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

- Network ACLs
- Security groups

PrivateLink and Lattice

- Getting started

rtb-065d968ea395aac31 / pb-routetable

Details [info](#)

Route table ID rtb-065d968ea395aac31	Main No	Explicit subnet associations subnet-0b95eb2689a105446 / pb-subnet	Edge associations -
VPC vpc-05d8c76cbaeb9e259 my-vpc	Owner ID 798974631771		

Routes Subnet associations Edge associations Route propagation Tags

Routes (2) [Filter routes](#) [Both](#) [Edit routes](#)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0488a0d97ecbefef5	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

7. Crate Route table for private subnet

- go to route table
- Create route table
- Use existing VPC
- No need to add routes.
- Associate subnet – edit subnet association – select private subnet and save.

VPC > Route tables > rtb-0c198224a76b7b6e4

Filter by VPC

Virtual private cloud

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- Route servers

Security

- Network ACLs
- Security groups

PrivateLink and Lattice

- Getting started

rtb-0c198224a76b7b6e4 / pv-routetable

Details [info](#)

Route table ID rtb-0c198224a76b7b6e4	Main No	Explicit subnet associations subnet-052e9fcc92210ef74 / pv-subnet	Edge associations -
VPC vpc-05d8c76cbaeb9e259 my-vpc	Owner ID 798974631771		

Routes Subnet associations Edge associations Route propagation Tags

Routes (1) [Filter routes](#) [Both](#) [Edit routes](#)

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

8. Crate Security Groups for each

Bastion Security Group

Name – *Bastion-sg*

Ibound- SSH(22)- assign to Your IP

Outbound- rule – All Traffic

☰ [VPC](#) > [Security Groups](#) > Create security group 🔍 🖨️ ⌂

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
SSH	TCP	22	My IP	
			111.125.244.2/32	

[Add rule](#) [Delete](#)

Outbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info
All traffic	All	All	Custom	
			0.0.0.0/0	

[Delete](#)

Private Ec2 Instance Security Group

Name – priavteec2-sg

Ibound- SSH(22)- assign to **Bastion-sg**

Outbound- rule – HTTP(80)- Assign to **LoadBalancer-sg**

☰ [VPC](#) > [Security Groups](#) > Create security group 🔍 🖨️ ⌂

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
SSH	TCP	22	Custom	
HTTP	TCP	80	Custom	

[Add rule](#) [Delete](#) [Delete](#)

Outbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info
All traffic	All	All	Custom	
			0.0.0.0/0	

[Delete](#)

Load Balancer Security Group

Name – loadbalncer-ig

Ibound- HTTP(80)- assign to Custom 0.0.0.0/0

Outbound- rule – HTTP(80)- Assign to **privateec2-sg**

VPC > Security Groups > Create security group

Description info
Loadbalancer Security Group

VPC info
vpc-02486129ea4c880ca

Inbound rules info

Type info	Protocol info	Port range info	Source info	Description - optional info	
HTTP	TCP	80	Anywh... 0.0.0.0/0		Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules info

Type info	Protocol info	Port range info	Destination info	Description - optional info	
HTTP	TCP	80	Custom sg-00705905d188cca0d		Delete

Add rule

Associate these security groups to Your VPC to avoid unnecessary Mistakes.

9. Launch Bastion Instance (Public EC2)

- EC2 → Launch instance
- Configure:
- AMI: Amazon Linux 2
- Instance Type: t2.micro
- Network: my-VPC
- Subnet: Public-Subnet
- Auto-assign Public IP: **Enabled**
- Security Group: Bastion-SG
- key pair
- Launch

key pair name - required
Proceed without a key pair (Not recommended) Default value Create new key pair

Network settings [Info](#) [Edit](#)

Network [Info](#)
vpc-02486129ea4c880ca

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☐ Create security group ☒ Select existing security group

Common security groups [Info](#)
Select security groups
Bastion-SG sg-064ebcd9bd55a0e7a [X](#)
VPC: vpc-02486129ea4c880ca

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023.9.2...[read more](#)
ami-068c0051b15c0b816

Virtual server type (instance type)
t2.micro

Firewall (security group)
Bastion-SG

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

10. Launch private Instance (web server)

- EC2 → Launch instance
- Configure:
- AMI: Amazon Linux 2
- Instance Type: t2.micro
- Network: my-VPC
- Subnet: Private-Subnet
- Auto-assign Public IP: **Disable**
- Security Group: *privateec2-sg*
- key pair
- Launch

Network settings [Info](#)

VPC - required [Info](#)
vpc-05d8c76c8aeb9e259 (my-vpc) [X](#)

Subnet [Info](#)
subnet-032e9f9c92210ef74 pv-subnet [X](#)
VPC: vpc-05d8c76c8aeb9e259 Owner: 798974631771
Availability Zone: us-east-1a (us-east-1a) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.2.0/24

Auto-assign public IP [Info](#)
Disable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☐ Create security group ☒ Select existing security group

Common security groups [Info](#)
Select security groups
private-ec2-sg sg-00705905d18cca0d [X](#)
VPC: vpc-02486129ea4c880ca

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

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Software Image (AMI)
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ami-068c0051b15c0b816

Virtual server type (instance type)
t3.micro

Firewall (security group)
private-ec2-sg

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

11. Now connect private instance to public by using commands:

SSH to Public Instance

```
ssh -i key.pem ec2-user@ <public_instance_ip>
```

Public Instance to Private Instance

```
ssh -i key.pem ec2-user@ <private_instance_ip>
```

```
#####
      _   _
     /_   _ \
    /  _ < /
   /  /___/
  /_____/

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Thu Dec 18 10:33:31 2025 from 34.201.2.139
[ec2-user@ip-10-0-1-163 ~]$ ssh -i "key.pem" ec2-user@34.201.2.139
#####
      _   _
     /_   _ \
    /  _ < /
   /  /___/
  /_____/

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Thu Dec 18 10:46:50 2025 from 18.206.107.28
[ec2-user@ip-10-0-1-163 ~]$
```

12. Install Apache Server on Private Instance

```
sudo yum update -y
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd
```

```
httpd-core-2.4.65-1.amzn2023.0.2.x86_64      httpdfilesystem-2.4.65-1.amzn2023.0.2.noarch      httpd-tools-2.4.65-1.amzn2023.0.2.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64      mailcap-2.1.49-3.amzn2023.0.3.noarch              mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.65-1.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-10-0-1-163 ~]$ sudo systemctl start httpd
sudo: systemctl: command not found
[ec2-user@ip-10-0-1-163 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-1-163 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-0-1-163 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Thu 2025-12-18 12:03:06 UTC; 30s ago
     Docs: man:httpd.service(8)
    Main PID: 6139 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
      Tasks: 177 (limit: 1067)
     Memory: 13.3M
        CPU: 5ms
    CGroup: /system.slice/httpd.service
            └─ 139 /usr/sbin/httpd -DFOREGROUND
              140 /usr/sbin/httpd -DFOREGROUND
              141 /usr/sbin/httpd -DFOREGROUND
              142 /usr/sbin/httpd -DFOREGROUND
              143 /usr/sbin/httpd -DFOREGROUND

Dec 18 12:03:06 ip-10-0-1-163.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Dec 18 12:03:06 ip-10-0-1-163.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server...
Dec 18 12:03:06 ip-10-0-1-163.ec2.internal httpd[6139]: Server configured, listening on: port 80
[ec2-user@ip-10-0-1-163 ~]$
```

13. Create Target Group

EC2- Target Groups – Create

EC2

Target groups

Create target group

Protocol

Protocol for communication between the load balancer and targets.

HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-04f3c513a22ec5d4 (my-task3-vpc)

10.0.0.0/16

Create VPC

Protocol version

☒ HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Step 1

Create target group

Step 2 - recommended

Register targets

Step 3

Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB

☐ IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB

☐ Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB

☐ Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 0/32

Protocol

Port

After Creating Target Group Must check status

Status Should be healthy

14. Create Application Load Balancer

EC2- Load balancer – Create

Select Application Load Balancer

Internet Facing

Custom VPC

Security Group

Listener- HTTP – Forward to target group

EC2 > Load balancers > Create Application Load Balancer

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

loadbalance-sg
sg-0ae06061740f56ca2 VPC: vpc-02486129ea4c880ca

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80 [Remove](#)

Protocol HTTP **Port** 80
1-65535

Default action [Info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

[Forward to target group](#) [Info](#)

How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

task3web-ALB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**

Includes only IPv4 addresses.

☐ **Dualstack**

Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **internet-facing** load balancers only.

After this

ALB will provide DNS – Paste that DNS into browser to access the website.

Website Hosted on Private EC2

Accessed via Load Balancer
